

Personnel: Who's on the Field Before Anything Else

Two digits, counted before the huddle breaks, tell you what the offense can do on this snap and force the defense's first answer.

First-and-10 near midfield, and the broadcast is still showing a replay of the last play. In the corner of the picture, a wide receiver jogs toward his sideline and a second tight end jogs out to the huddle. On the far side of the field a defensive coach shouts something with a number in it. A cornerback peels away from his teammates and runs to the bench, and a linebacker runs out to take his place. Nobody has lined up yet. The quarterback hasn't even said the play. But both teams have already made a move, and if you were counting, you have already learned something about the next snap. This chapter is about that count: who the offense sends onto the field, the two-digit code coaches use for it, why it is the very first clue of every play, and why the defense is allowed to answer it.

You'll be able to...

- Decode any personnel number (11, 12, 21, 13, 22, 10, 20, 00) into running backs, tight ends and wide receivers, and count it yourself from the couch before the snap.
- Explain why personnel is the first pre-snap clue, and how it pushes the defense into a substitution of its own.
- State the substitution rule (when the offense changes players, the defense gets time to match) and say what an offense gains by *not* substituting.
- Read the league's personnel trends from 2016 to 2025 and explain why 11 personnel fell to a decade low in 2025 while 12 personnel hit a decade high.
- Use real pass rates by personnel to make a run-or-pass lean, and say why it's a lean, not a certainty.

Eleven players, five choices

Start with the arithmetic, because the whole idea falls out of it. An offense has eleven players. Six of them are almost never in question: the five **offensive linemen** (center, two guards, two tackles) and the **quarterback**. The rules require at least seven players on the line of scrimmage, and the five linemen are the core of that seven; the quarterback runs the play. That leaves **five spots**, and those five are where the coach chooses. He fills them from three kinds of player you met in The Twenty-Two:

- **Running backs**, including **fullbacks**: ball carriers, lead blockers and pass protectors who line up behind the line.
- **Tight ends**: big enough to block a defensive end, fast enough to run a route. The hybrid.
- **Wide receivers**: the fast, smaller players who stretch the field.

A **personnel grouping** is the answer to "which mix of those three is on the field?", and NFL coaches write it as two digits. **The first digit is the number of running backs. The second is the number of tight ends.** You never say the receivers, because they're whatever is left:

$$\text{wide receivers} = 5 - \text{running backs} - \text{tight ends}$$

So "11" (say it "one-one," or "eleven") is one back and one tight end, which leaves three receivers. "12" ("one-two") is one back, two tight ends, two receivers. "21" is two backs, one tight end, two receivers. Nobody is quite sure who first wrote personnel this way; it's coaches' shorthand that became the league's common language, and the NFL's own Next Gen Stats now uses it.¹ The code counts backs and tight ends, not receivers, because they are the players that change the offense's *character*. A third receiver is one more pass catcher. A second tight end or a fullback is a decision to block and run, or at least to look as if you might.

¹For the code as the league's own vocabulary, see e.g. NFL.com, "NFL trend watch: Heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks" (Next Gen Stats), which reports usage by grouping: <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>. Some staffs also use words for groupings (for example "Regular" for two backs, one tight end and two receivers).

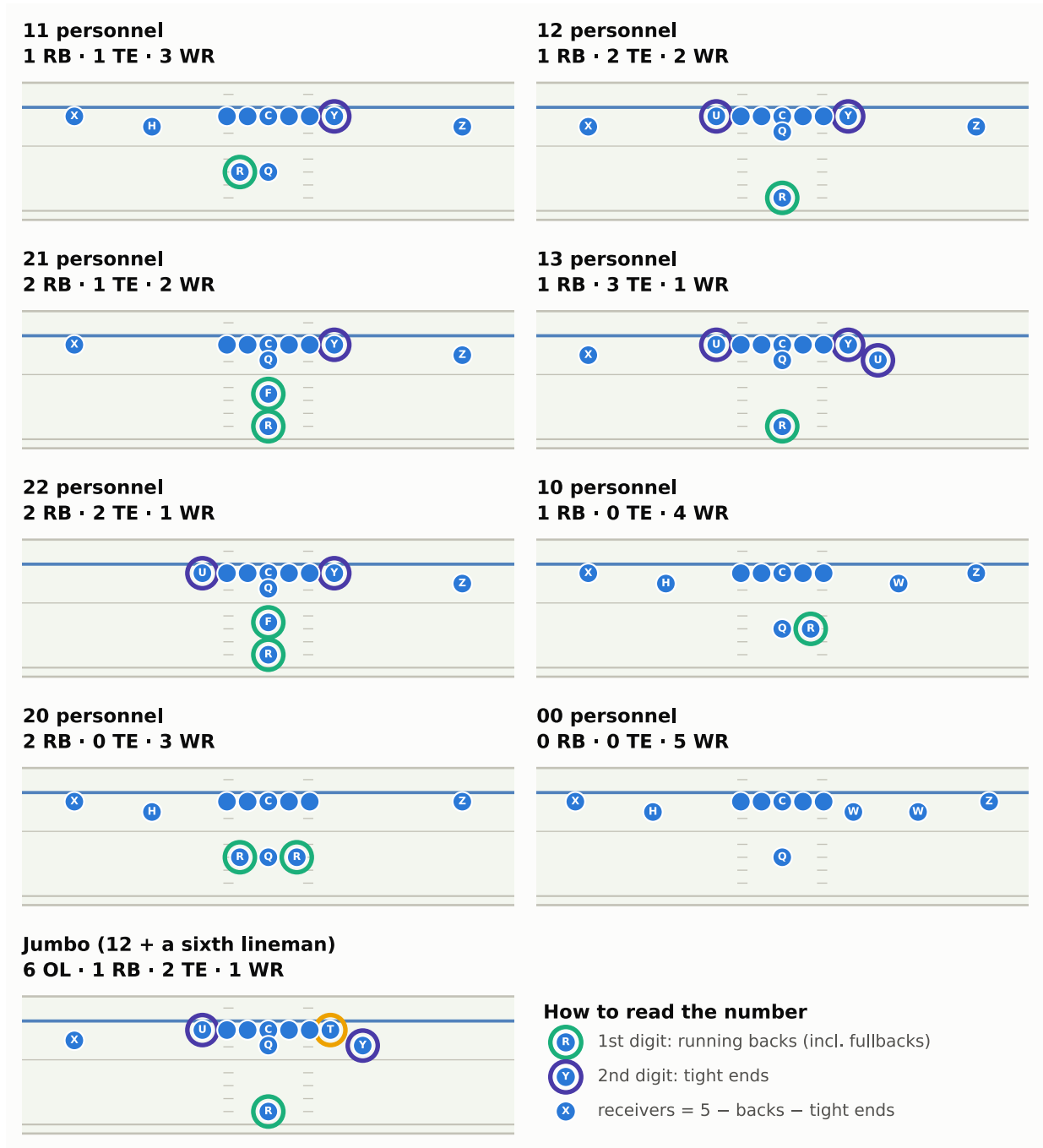


Figure 1: The personnel card deck: one typical alignment for each grouping. Every plate has the same five linemen and quarterback; only the five rings and blue receivers change. Count the aqua rings (running backs, including fullbacks) for the first digit and the purple rings (tight ends) for the second; whatever is left are wide receivers. Jumbo breaks the arithmetic: a sixth lineman (gold ring) takes a skill player's place, so '12' there means one receiver, not two. Letters: Q quarterback, R running back, F fullback, Y and U tight ends, X, Z, H and W wide receivers. These are typical looks only; each grouping can line up in many formations.

The zeros, and the one that breaks the rule

Three groupings in the deck have a zero in them, and they read the same way. “10” is one back and *zero* tight ends, so four receivers. “20” is two backs and no tight end: three receivers, with two backs beside the quarterback, a look built to threaten a run either way or to send both backs out as receivers. “00” (“zero-zero,” or just **empty**) is no backs and no tight ends: five receivers spread across the field and a quarterback alone in the backfield. You'll also see a 01 now and then (no backs, one tight end, four receivers), and on goal-line snaps a 23 or a 32. The rule never changes.

The rule *does* break in one place. The arithmetic assumes five linemen. When a coach puts a **sixth** offensive lineman on the field, usually a backup tackle lined up where a tight end would stand, the code keeps counting backs and tight ends but there's one

fewer receiver than it promises. That's **jumbo**: in nflverse's data it shows up as a string like "6 OL, 1 RB, 2 TE, 1 WR," which is 12 personnel by the digits but only one receiver on the field. Teams use it where a yard matters more than a pass catcher: third-and-1, fourth-and-inches, the goal line. The extra lineman can catch a pass only if he tells the referee before the snap that he is reporting eligible (the referee then announces it to the defense); the rules of eligibility and the tricks teams play with them belong to the next chapter.

Heavy personnel is the umbrella term for everything with fewer than three receivers: 12, 13, 21, 22, jumbo and the rest. "Light" or "spread" personnel is the other end: 11, 10, 00.

Counting from the couch

The count is easier than it sounds, and you can do it in the eight or so seconds between the huddle breaking and the snap. Work in this order:

1. **Backs.** Look behind the quarterback. One player beside or behind him is one back. A second player directly in front of the deep one (an I) or beside the quarterback on the other side is a second back. A fullback is a back.
2. **Tight ends.** Look at the ends of the offensive line. A big player standing or crouched right next to a tackle is almost always a tight end. So is a big player just behind and outside that spot.
3. **Subtract.** Everyone else is a receiver. You don't need to find them.

Two traps. First, **position is who a player is, not where he stands**. A tight end split out wide is still a tight end; a running back lined up in the slot is still a back. The count is about bodies, not spots. Second, jersey numbers have stopped helping: since the 2021 numbering change, tight ends and receivers can wear many of the same numbers (The Twenty-Two has the details). Size and stance are better tells: tight ends are usually 245–260 pounds and look it; receivers are often 60 pounds lighter. Broadcasters help too. When the analyst says "they've gone to 12 personnel," check your count against his.

i Madden vs. real life

In Madden you pick a formation and the game quietly puts the right players in it; the defense's substitutions happen instantly and invisibly. In the NFL, personnel is a separate, earlier decision. The offensive play-caller's first words on the radio are often the personnel group, so the right substitutes can start running onto the field before the rest of the call is even spoken (that order is part of the chain in What a Play Really Is). The defense watches those substitutes and answers with its own. All of that happens *before* the formation, before the play, and in full view of the TV camera, if you know to look.

Why personnel comes first

Every snap is a series of reveals. The offense shows its personnel first, as players run on and off. Then its formation, when it breaks the huddle and lines up. Then any motion or shift. Then the play itself, at the snap. Personnel is the earliest reveal, which is why it's the first clue you should collect.

It matters because **personnel limits the menu**. A group with two tight ends and a fullback can run a power play with three extra blockers at the point of attack, but it can't put four receivers out wide. A group with four receivers can stretch a defense across all 53 yards but has only five linemen and a back to block a run. Within a grouping the offense still has hundreds of plays, but it has given away which shelf of the playbook they come from. So the offense is always trading two things against each other:

- **Capability.** More receivers means more ways to attack in the passing game. More tight ends and backs means more blockers, a stronger run game, and better protection for play-action passes.
- **Information.** Every grouping tells the defense something, and the defense gets to act on it, because it can change its own players in response.

That second half is the reason personnel is a contest, not just a choice. The defense sorts its own players into **defensive packages**: **base defense** has four defensive backs (two cornerbacks, two safeties) and three or four linebackers; **nickel** swaps a linebacker for a fifth defensive back, usually a slot cornerback; **dime** has six. (Those packages, and why nickel became the modern default, are taught properly in A First Look at Defense.) The rough rule of **substitution matching** is simple: meet receivers with defensive backs, and meet tight ends and backs with linebackers and linemen. More receivers, more cover men. More big bodies, more big bodies.

! Why it exists: personnel groupings

Before 1950, NFL rules restricted substitution so tightly that most players played both offense and defense, and a coach had little reason to name his skill-position mixes. On January 20, 1950, the league restored unlimited free substitution, which opened the era of two platoons and specialists.² Once a coach could send in any player on any down, the question “which five skill players for this snap?” became a weapon, and the defense’s mirror question, “which five or six defensive backs?”, became the answer. The two-digit code is just the efficient way to say it in a headset with the play clock running.

The substitution rule: the defense gets to answer

If the offense could change players and snap the ball instantly, the defense would always be guessing. The rulebook stops that. Before the two-minute warning of either half, when the offense sends in substitutes, it **may not snap the ball until the defense has had a chance to respond**. The umpire (the official who stands among the defenders) moves over the ball and stays there until the referee judges that the defense has had a reasonable interval to make its own changes.³ If the offense snaps over him, or tries to, it’s a five-yard penalty. You can see this happen every week: an official standing over the football while the play clock runs, then trotting back to his spot.

The protection runs one way only. If the offense *doesn’t* substitute, there is no pause, and a defense that changes players anyway risks being caught with twelve men on the field or in the wrong package when the ball is snapped (What a Play Really Is covers the twelve-men penalty and the “free play” it can create). That one-way protection explains three things you’ll see on Sundays:

- **The defense waits for the offense.** Defensive substitutes stand on the sideline watching the offensive substitutes. The defensive coach shouts the matching package only after he sees who ran on.
- **No-huddle offenses love keeping the same eleven.** With no substitution, there is no umpire pause, so the defense is stuck with whatever package it had on the field. That’s a large part of why tempo works, and why an offense with versatile players (a tight end who can also play like a receiver) can go fast without giving up matchups.
- **Defenses learned to game it, and the league pushed back.** By 2025 some defenses were deliberately slow to complete their swaps, burning the offense’s play clock. In October 2025 officials were told to signal a formal substitution window when the offense subs during a running play clock, and to close it if the defense isn’t substituting at a normal speed, even with twelve defenders still on the field. The other way round, an offensive player who runs on and straight back off to fake a substitution draws a warning the first time and a 15-yard unsportsmanlike-conduct penalty after that. Field goals and the last two minutes of each half are excepted.⁴

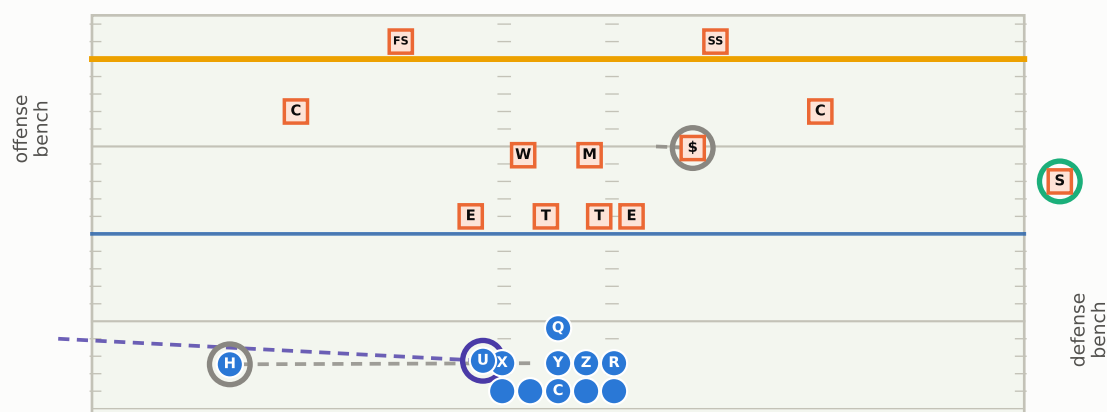
The figure walks through the most common swap in football: the offense trades its third receiver for a second tight end (11 personnel becomes 12), and the defense answers by trading its fifth defensive back for a third linebacker (nickel becomes base).

²Pro Football Hall of Fame, NFL history chronology, 1950: “Unlimited free substitution was restored on January 20, opening the way for the era of two platoons and specialization in pro football”: <https://www.profootballhof.com/football-history/football-history/1940-1959/1950>

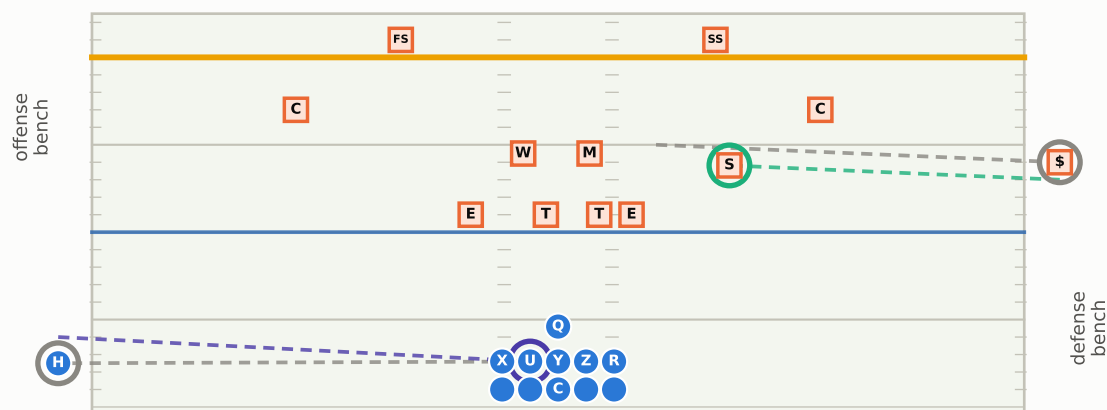
³NFL Rulebook, Rule 5, Section 2 (substitutes), Article 10 in the 2026 edition: before the two-minute warning of either half, when the offense substitutes, it must not snap until the defense has been permitted to respond; the umpire stands over the ball until the referee judges the defense has had a reasonable interval: <https://operations.nfl.com/the-rules/nfl-rulebook/>

⁴Football Zebras, “NFL officials instructed on new procedures for substitution tactics” (Oct. 2, 2025), on the “iron cross” signal that opens a substitution period, the instruction to close it if the defense is not substituting at a normal speed, the warning and then 15-yard unsportsmanlike-conduct penalty for simulated offensive substitutions, and the exceptions for field goals and the two-minute periods: <https://www.footballzebras.com/2025/10/nfl-officials-instructed-on-new-procedures-for-substitution-tactics/>

:34 — offense swaps a receiver (H) for a tight end (U)



:29 — defense answers: nickel back (\$) off, linebacker (S) on



:18 — lined up: 12 personnel against base

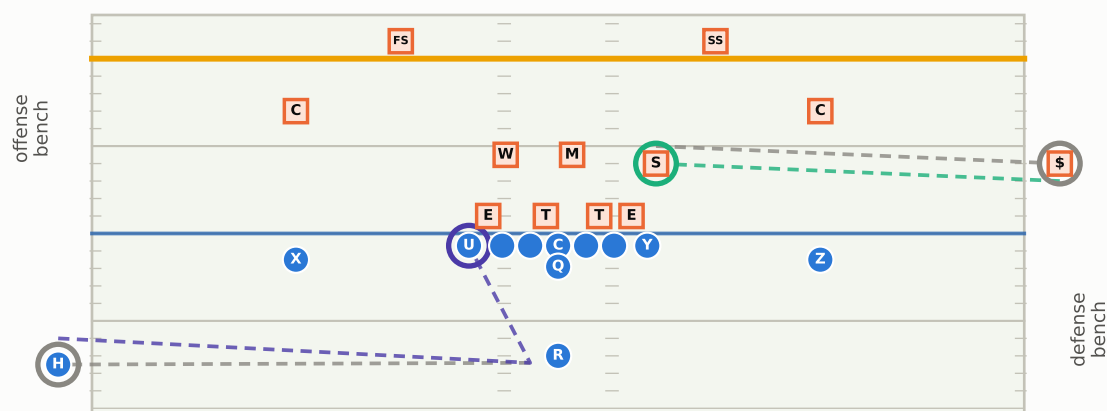


Figure 2: The substitution game, simulated (play clock times are typical). Top: the offense, huddled in 11 personnel, sends H (a wide receiver) to its bench and brings on U, a second tight end: 12 personnel. Middle: the defense sees two tight ends and answers. Its nickel back (\$) jogs to the defensive bench, on the opposite sideline, and a third linebacker (S, the Sam) runs on. The umpire holds the snap until this is done. Bottom: both sides line up, 12 personnel against a base 4-3. Dashed lines trace the four substitutes.

Two details in that picture are easy to miss on TV. The benches are on **opposite sidelines**, so the offensive and defensive substitutes run in opposite directions; if you only watch one sideline, you'll see half the exchange. And the defense moves **second**, on purpose. A defensive coordinator who guesses early and gets it wrong is stuck: the offense can then switch again, and the officials will let the defense answer again, but every exchange eats the play clock that the defense's own play call needs.

Common misconception

“The defense has to match.” It doesn’t. Matching is a choice, and good defenses break the rule on purpose. Some keep nickel on the field against two tight ends because their slot cornerback tackles like a linebacker. Some play a “big nickel” with a third safety instead of a corner. Some refuse to substitute against a fast no-huddle team because changing players is exactly what the offense wants them to try. The substitution rule guarantees the defense the *chance* to match, not the obligation. What the defense gives up when it doesn’t match is the subject of the next section.

The matchup problem: why 12 personnel is a puzzle

Personnel is a contest because the offense can do something after the defense has answered: **line up**. The defense picked its players for the personnel it saw. The offense then picks the formation, and it can put those players wherever it likes. That’s the heart of the matter, and 12 personnel shows it most clearly.

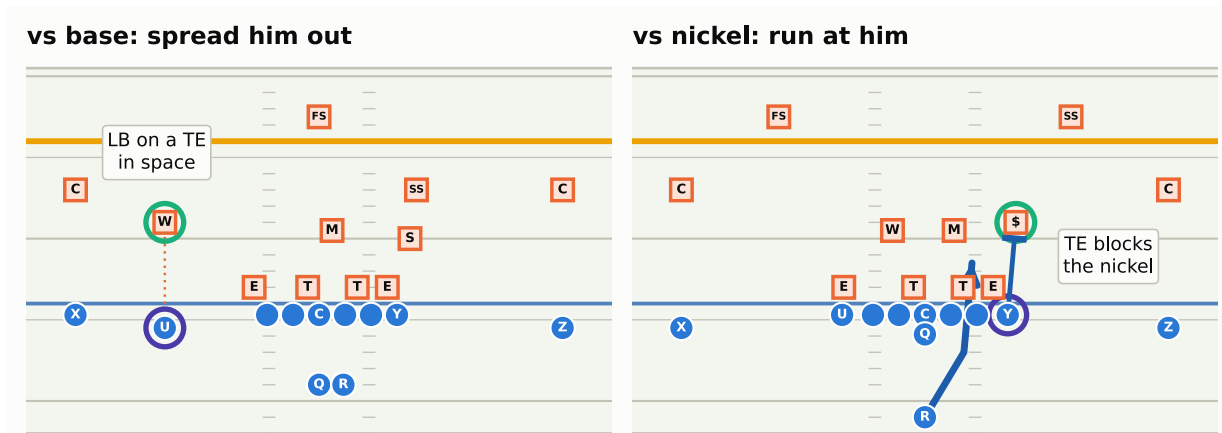


Figure 3: The 12-personnel puzzle. Left: the defense matched two tight ends with base (three linebackers, four defensive backs), so the offense splits a tight end (U) out into the slot. Now a linebacker (W) has to cover him in open space, the matchup the offense wanted. Right: the defense chose nickel instead (five defensive backs), so the offense keeps both tight ends in tight and runs the ball. Now the nickel back (\$), often 30 to 50 pounds lighter than a linebacker, has to take on a tight end’s block. Same eleven offensive players; the defense picks which problem it would rather have.

Against **base**, the defense has bigger, slower players. The offense spreads them out: flex a tight end into the slot and a linebacker must cover him, or put a back out wide and see who follows. Against **nickel**, the defense has a smaller player where a linebacker used to be. The offense keeps the tight ends in tight and runs right at him. Either answer leaves a weakness, and the offense gets to choose which one to attack *after* seeing the answer. Football people call this “playing 11 out of 12”: a 12-personnel group whose tight ends can block like linemen and catch like receivers forces the defense to be wrong one way or the other.

i Film room: Patriots 2011

The most famous version of this puzzle belonged to New England in 2011. Rob Gronkowski and Aaron Hernandez were both tight ends, and both could line up anywhere. When defenses brought on linebackers to match two tight ends, the Patriots split one or both out wide and threw against linebackers. When defenses went to nickel, they put both in tight and ran. And because the same five skill players could play both ways, the Patriots could go no-huddle without substituting, which froze the defense in whatever it had guessed. Contemporary coverage summed up the defensive coordinator’s choice the way the figure above does: whatever he picked would probably be wrong. Gronkowski set the tight-end record for receiving touchdowns that season.⁵ What to notice: the problem wasn’t two tight ends. It was two tight ends the defense could not label as either blockers or receivers.

⁵NFL.com, “Dynamic TE duo is secret behind Pats’ offensive explosion” (2011): <https://www.nfl.com/news/dynamic-te-duo-is-secret-behind-pats-offensive-explosion-09000d5d8226c4ae>; Associated Press via *The Washington Times*, “Patriots’ 2 tight ends pose problems for foes” (Sept. 14, 2011): <https://www.washingtontimes.com/news/2011/sep/14/patriots-2-tight-ends-pose-problems-for-foes/>. Gronkowski’s 17 receiving touchdowns in 2011 set the record for a tight end.

How defenses actually answer

The data shows substitution matching happening, and also shows its limits. In the 2025 nflverse participation data, defenses answered 11 personnel with nickel or dime on more than nine snaps in ten. Against 12 and 21 they leaned to base, but only about three times in five; roughly a third of the time they stayed in nickel and accepted the run-game risk. Against 13 and 22, the heaviest groups, base was close to automatic.⁶

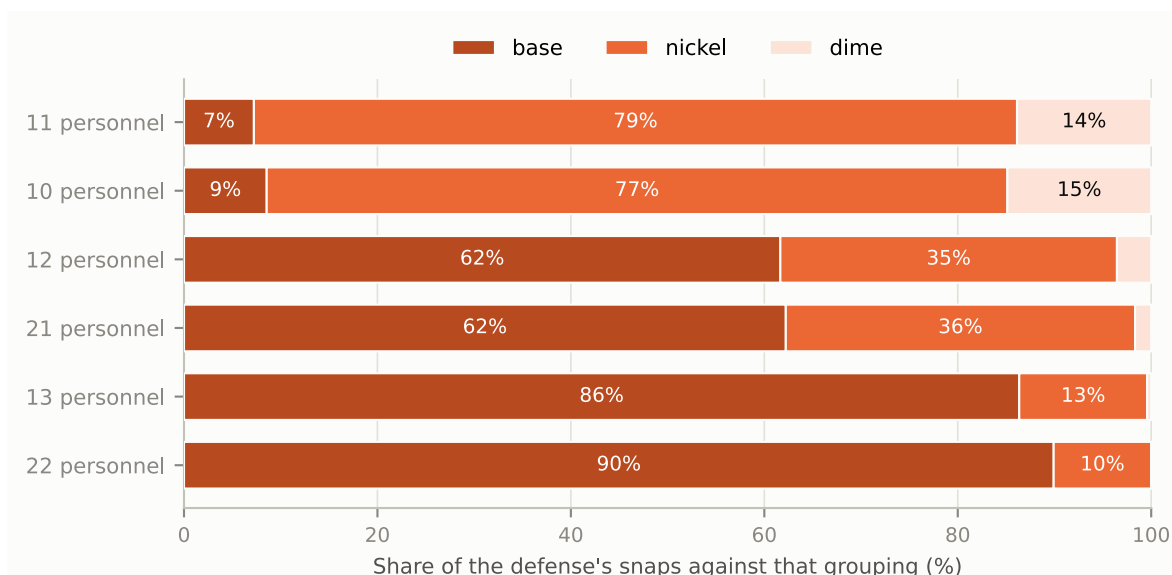


Figure 4: How defenses answered each offensive grouping, 2025 regular season (runs and passes). ‘Base’ here means four or fewer defensive backs, ‘nickel’ five, ‘dime’ six or more. More receivers bring more defensive backs, but against 12 and 21 personnel about a third of defenses stayed in nickel anyway. Source: nflverse participation data (FTN charting); defensive backs counted by roster position, so a safety playing linebacker still counts as a defensive back.

That middle row is where the chess game lives. A defense that stays in nickel against 12 personnel is betting that its slot cornerback can survive in the run game, and that the offense will throw. A defense that goes to base is betting its linebackers can cover. Neither bet is free, and the best offenses build their week around finding out which bet a defense likes to make.

What personnel tells you about run and pass

Here’s the payoff for the coach. If personnel limits the offense’s menu, it should tilt the odds of a run or a pass, and it does. The chart uses three seasons of plays on first and second down in **neutral situations**, meaning not late and not lopsided: outside the last two minutes of each half and with neither team’s win probability above 80%. Those filters remove the snaps where the score or the clock, not the offense’s preference, decides what gets called. (Analysts formalize this idea in Tendencies.)

⁶Course calculation from nflverse `load_participation()` (FTN source) joined to play-by-play, 2025 regular season, runs and passes with eleven defenders listed. Base = four or fewer defensive backs by roster position (CB, S, FS, SS, DB), nickel = five, dime = six or more. Against 11 personnel: nickel 78.9%, dime 13.9%, base 7.2%. Against 12: base 61.7%, nickel 34.8%. Against 21: base 62.2%, nickel 36.1%. Against 13: base 86.4%. Against 22: base 89.9%. Because the FTN strings list roster positions, a safety used as a linebacker counts as a defensive back, so “base” here is a lower bound.

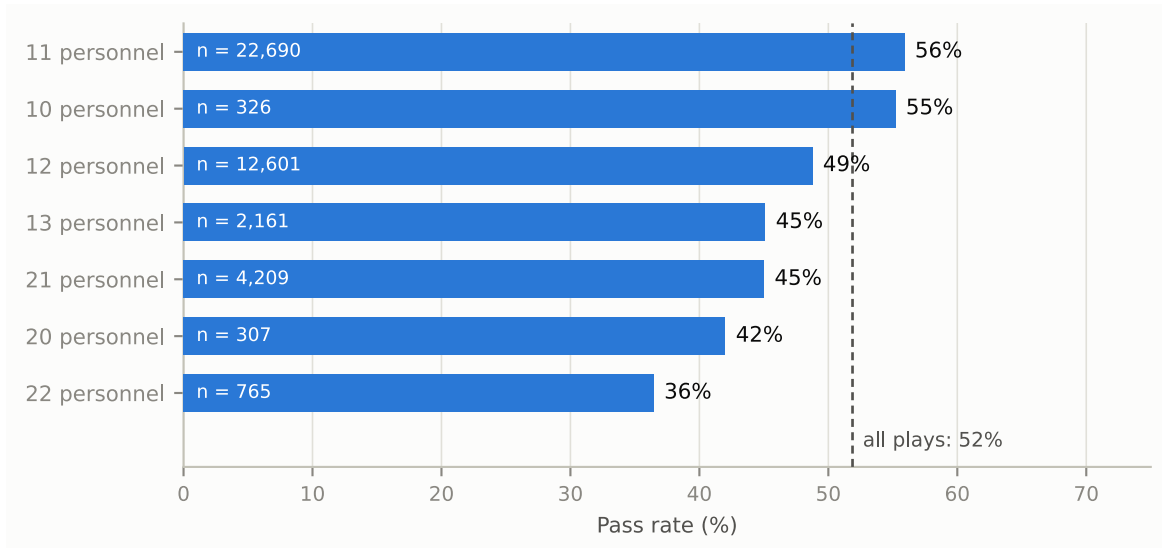


Figure 5: How often each personnel grouping dropped back to pass, first and second down in neutral situations (win probability 20–80%, outside the last two minutes of each half), 2023–2025 regular seasons. Dropbacks include sacks and scrambles. Groupings with fewer than 300 such plays are left out. Each bar is labelled with its pass rate and its number of plays. The tilt is real but gentle: no common grouping is a sure run or a sure pass. Source: nflverse play-by-play joined to participation data.

Read the chart from the bottom up. The groups with more receivers pass more: 11 and 10 personnel drop back on a little more than half of these snaps. The heavier you go, the more the offense runs: 12 personnel is close to a coin flip, 21 and 13 lean run, and 22 runs about two times in three.⁷ So personnel gives you a real lean, and you should use it. But look at the size of the lean. On early downs in a neutral game, even 22 personnel passes about a third of the time, and 11 personnel runs more than four times in ten. Personnel is a tilt, not a tell.

Why isn't it stronger? Because the offense knows the defense is reading the same chart. If 12 personnel always ran, defenses would always answer with base and crowd the line, and the offense's best plays from 12 would be **play-action passes**, fakes to the running back designed to pull defenders forward (Play-Action, Bootlegs, and Screens). Offenses spend their heavy personnel on passes often enough to keep defenses honest, and their light personnel on runs often enough to punish a defense that drops everyone into coverage. The groupings that *are* lopsided tend to be the ones that show up only in lopsided situations.

🔥 Common misconception

"Two tight ends means run." On early downs in close games, NFL offenses in 12 personnel threw the ball almost half the time from 2023 to 2025. The groupings that really do predict a run are the rare, very heavy ones (22, jumbo, goal-line sets with three tight ends and two backs), and even those throw often enough near the goal line to punish a defense that sells out.

Down and distance change the mix

Personnel also follows the situation, which is why it's best read *together* with the down-and-distance graphic you learned in Downs, Distance, and Field Position. In 2025, offenses used 11 personnel on about 54% of first and second downs, but on 86% of third and fourth downs with 7 or more yards to go. On third and fourth down with 2 yards or less, 11 fell to about half, and 12, 13 and 22 together made up about 40%.⁸ With the ball at the 1- or 2-yard line, 12 personnel was the single most common choice, and almost a third of those plays had six or more offensive linemen on the field.

So a better habit than "what does this personnel mean?" is "is this personnel *normal* for this down and distance?" Eleven personnel on third-and-8 tells you almost nothing; everyone does it. Two tight ends and a fullback on third-and-8 would tell you a lot.

⁷Course calculation from nflverse play-by-play joined to participation data, 2023–2025 regular seasons, first and second down, win probability 20–80%, more than two minutes left in the half; pass = dropback (`pass == 1`, including sacks and scrambles). Pass rates: 22 personnel 36.5%, 21 45.0%, 13 45.1%, 12 48.8%, 10 55.2%, 11 55.9%. Personnel = running backs (RB + FB) followed by tight ends.

⁸Course calculation from nflverse play-by-play joined to participation data, 2025 regular season, runs and passes. 11 personnel: 53.6% of first and second downs; 85.7% of third and fourth downs with 7+ yards to go; 51.5% of third and fourth downs with 2 or fewer (12 personnel 24.1%, 13 8.6%, 22 7.4%). With the ball at the opponent's 1 or 2: 12 personnel 33.1%, 11 28.2%, 13 16.9%, 22 10.8%; plays with six or more offensive linemen 30.7%.

Personnel means different things for different teams

The chart averages over 32 offenses, and personnel habits are a team's fingerprint. San Francisco has built its offense around 21 personnel with fullback Kyle Juszyk for most of Kyle Shanahan's tenure, and it has used 21 more than any team, or second-most, in every season from 2109 through 2025, roughly a third of its snaps when the league average is about 7%. What makes the 49ers' version dangerous is that they *throw* from it. On neutral early downs from 2023 to 2025, San Francisco passed on 53–54% of its 21-personnel snaps; the rest of the league, in the same grouping, passed about 42–46%.⁹ A defense that reads "fullback on the field, expect the run" is reading the league's tendency, not this team's. That's why coaches scout each opponent's personnel tendencies every week (The Game Plan), and why your own read gets sharper the more you watch one team.

A decade of personnel: the rise of 11 and the heavy comeback

Personnel is also the clearest window onto how offenses have changed. The chart shows the share of each common grouping across the league since 2016.

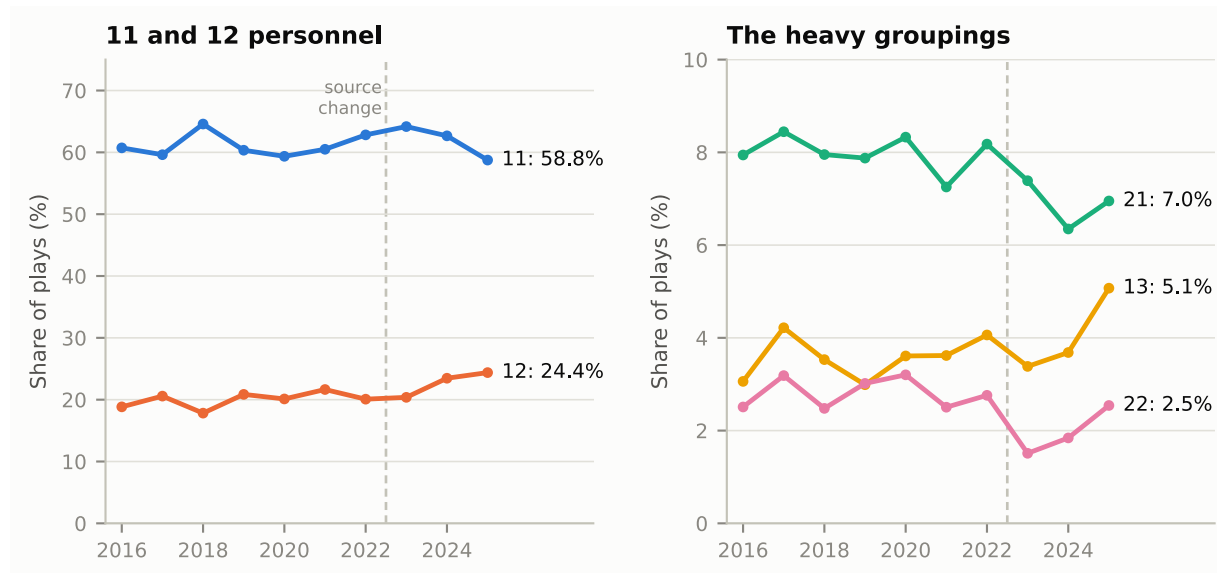


Figure 6: League-wide personnel share by season, 2016–2025 (regular season, runs and passes). Left: the two big groupings. 11 personnel dominated for a decade, then fell to 58.8% in 2025, its lowest share in the series, while 12 personnel rose to 24.4%, its highest. Right: the heavy groupings on their own scale; 13 personnel also reached a decade high in 2025. The dashed line marks the change in data source between 2022 (NGS) and 2023 (FTN charting). Source: nflverse participation joined to play-by-play.

Three stories are layered in those lines.

First, 11 personnel became the NFL's real base offense. For most of football history the default was two backs. Then offenses spread out, through the 1990s and 2000s, borrowing from college spread systems, and pass rules that favoured receivers made a third wide receiver more valuable than a fullback. By the start of this chart, 11 was already on about 60% of snaps, and defenses had answered by making nickel their default too.

Second, the extreme versions taught the league something. The 2017–2019 Rams under Sean McVay made 11 personnel almost their only grouping, and the payoff came in the run game, against defenses that had to stay small (Film room, below). At the other end, the 2019 Ravens built a record-setting rushing offense on heavy sets around quarterback Lamar Jackson. Offenses learned that being *unusual* has value of its own: a defense that rarely practices against 22 personnel is worse at defending it.

Third, the pendulum swung back. By 2025, defenses were lighter than ever: nickel on most snaps, and two deep safeties more often than at any point since tracking began (that shift is the story of A First Look at Defense and Part 6). Offenses answered the way the dilemma figure predicts: if the defense is going to play small, put big people on the field and run at it. In 2025, 11 personnel fell to 58.8% of snaps, its lowest share in the ten seasons charted here, and 12 personnel climbed to 24.4%, its highest. Thirteen

⁹Course calculation from nflverse play-by-play joined to participation data. San Francisco's 21-personnel share and league rank: 27.2% (1st) in 2019, 33.4% (2nd) 2020, 36.2% (1st) 2021, 32.8% (2nd) 2022, 37.2% (2nd) 2023, 35.6% (1st) 2024, 37.1% (1st) 2025; the league share of 21 personnel was 6.3–8.4% in those seasons. Pass rate from 21 on neutral early downs (as in the pass-rate chart): SF 54.1% (2023), 54.1% (2024), 53.0% (2025); all other teams 46.0%, 41.5%, 42.0%.

personnel rose to 5.1%, also a decade high, and plays with six or more offensive linemen nearly doubled, to about 5%.¹⁰ Next Gen Stats, measuring a little differently, found offenses used fewer than three wide receivers on 41.7% of 2025 plays, the first season at or above 38% since its data begin in 2016, with the rate climbing during the season, from 38.8% in Weeks 1–9 to 44.5% after.¹¹

And the defense answered back. The same Next Gen Stats report found defenses in base on 29.7% of 2025 snaps, up from a low of 20.5% in 2023. Personnel is an arms race in miniature: each side's habits create the other side's opportunity, and the opportunity, once exploited, changes the habits. The long version of that arms race is The Modern Era.

i Film room: Rams 2018

Sean McVay's second Rams team used 11 personnel on 92.7% of its snaps in 2018, the highest share in the league by a wide margin (the next team was under 76%), on the way to Super Bowl LIII. Why commit to one grouping? Because it took away the defense's best weapon against the offense: information. With the same three receivers, tight end and back on almost every snap, the Rams' personnel told the defense nothing, so defenses stayed in nickel, and the Rams ran the ball at those lighter fronts. In 2018, 45% of the Rams' runs came against six or fewer defenders near the line (league: 37%), and their runs averaged a positive EPA while the league's averaged a negative one.¹² (EPA, expected points added, is the per-play scorecard from Downs, Distance, and Field Position.) What to notice: the Rams hid their intentions in their *formations and motion*, not their personnel, which is exactly the pre-snap game the next chapters teach.

i Film room: Ravens 2019 and Eagles 2024

Two championship-level offenses went the other way. The 2019 Ravens used 22 personnel (two backs, two tight ends) on 16.5% of their snaps, more than five times the league rate and the most of any team, and 11 personnel on under half; they ran for 3,296 yards, an NFL single-season team record.¹³ Five seasons later the Eagles, with Saquon Barkley, went from 12 personnel on 21.6% of snaps in 2023 to 33.3% in 2024, and ran the ball on 70% of their 12-personnel plays. Barkley ran for 2,005 yards and the Eagles won Super Bowl LIX.¹⁴ What to notice: both offenses *announced* the run with their personnel and won anyway, because their run game was good enough that knowing it was coming didn't help. Heavy personnel is a bet on your blockers.

Jumbo and the eligible tackle

The heaviest corner of the chart is the sixth lineman. Under Ben Johnson, offensive coordinator from 2022 to 2024, the Detroit Lions were among the league's most frequent jumbo teams, with backup tackle Dan Skipper the usual sixth man, often reporting eligible so that the defense had to account for him as a receiver.¹⁵ It produced one of the strangest finishes of the 2023 season. On December 30, 2023, trailing Dallas 20–19 in the final minute, the Lions scored a touchdown, went for two, and threw to left tackle Taylor Decker, who caught it. The officials ruled him ineligible: they had heard Skipper, not Decker, report. The Lions said Decker had reported and Skipper had not; the referee's pool report said the reverse. The catch came off the board, Detroit's third

¹⁰Course calculation from nflverse play-by-play joined to `load_participation()`, regular seasons 2016–2025, plays with `play_type` pass or run and a recorded down; personnel = RB + FB count followed by TE count. 11 personnel: 60.7% (2016) ... 64.6% (2018), 64.2% (2023), 58.8% (2025). 12 personnel: 18.8% (2016) ... 24.4% (2025). 13 personnel: 3.1% (2016) ... 5.1% (2025). Plays with six or more offensive linemen: 2.6–3.1% in 2022–2024, 5.4% in 2025. Participation comes from NGS for 2016–2022 and FTN from 2023; the two sources format personnel differently (FTN lists roster positions), and the code above normalizes both, but cross-break comparisons, especially of lineman counts, carry that caveat. These figures match the course fact sheet (`planning/FACTS-current.md`, §10).

¹¹NFL.com, “NFL trend watch: Heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks” (Next Gen Stats, 2026): fewer than three wide receivers on 41.7% of 2025 plays; 38.8% in Weeks 1–9 and 44.5% from Week 10; defenses in base on 29.7% of 2025 snaps against a low of 20.5% in 2023: <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>. NGS's series is measured differently from the nflverse calculation in this chapter, so the two are quoted side by side, never on one chart.

¹²Course calculation from nflverse play-by-play joined to participation data. Rams 11-personnel share: 81.4% in 2017 (1st; league 59.6%), 92.7% in 2018 (1st; next highest Green Bay 75.9%; league 64.6%), 73.0% in 2019 (4th). 2018 runs: 44.8% against six or fewer in the box (`defenders_in_box`, league 36.7%); mean EPA per run +0.07 (league –0.07). Super Bowl LIII: Patriots 13, Rams 3 (Feb. 3, 2019).

¹³Course calculation from nflverse play-by-play joined to participation data: Baltimore 2019, 22 personnel 16.5% of snaps (next Las Vegas 12.3%; league 3.0%), 11 personnel 45.5%, fewer than three wide receivers 52.8%. Team rushing record (3,296 yards): Pro Football Reference, 2019 Baltimore Ravens: <https://www.pro-football-reference.com/teams/rav/2019.htm>

¹⁴Course calculation from nflverse play-by-play joined to participation data: Philadelphia 12 personnel 21.6% of snaps in 2023 (15th) and 33.3% in 2024 (6th); runs on 69.9% of its 2024 12-personnel plays (all situations). Barkley's 2,005 rushing yards (2024): Pro Football Reference: <https://www.pro-football-reference.com/players/B/BarkSa00.htm>. Super Bowl LIX: Eagles 40, Chiefs 22 (course fact sheet, §1).

¹⁵Course calculation from nflverse participation data, share of runs and passes with six or more offensive linemen: Detroit 10.9% in 2021 and 9.8% in 2022 (2nd in the league both years), 5.1% in 2023 (7th) and 6.4% in 2024 (6th). Lineman counts come from NGS strings through 2022 and FTN roster positions from 2023, so the two periods are not strictly comparable. Ben Johnson was Detroit's offensive coordinator 2022–2024 (course fact sheet, §0 C1). On Skipper's role:

try at two points fell incomplete, and the Lions lost 20–19.¹⁶ The rules of reporting, and why the league won’t loosen them, are in Formation Rules and the Language of Alignment; the personnel lesson is simpler. Once the offense brings on a sixth lineman, the defense has to find out *which* of the big men can catch a pass, and the referee’s announcement is the only way to know.

💡 Watch for it on Sunday: call the personnel

For one full quarter (two if you can), do this before every snap.

- **Count and say it.** Backs, then tight ends, out loud: “one-one,” “one-two,” “two-one.” Check your count when the broadcast analyst mentions personnel, or on the replay’s wide shot.
- **Catch the swap.** Between plays, watch the offense’s sideline. When a receiver runs off and a tight end or fullback runs on, glance at the defense’s sideline: who comes off? Did it match?
- **Spot the umpire.** If the offense substituted, look for the official standing over the ball. When he jogs away, the defense has had its chance.
- **Make the lean.** Before the snap, guess run or pass using the personnel chart above *and* the down and distance. Then ask: is this personnel normal for this down?
- **Log it.** Add a “personnel” column to the prediction log you started in How to Watch a Broadcast Like You Mean It. After the game, compare your hit rate on 11-personnel snaps with your rate on heavy ones.

Predict the play

💡 Predict the play 1: count it, then find the problem

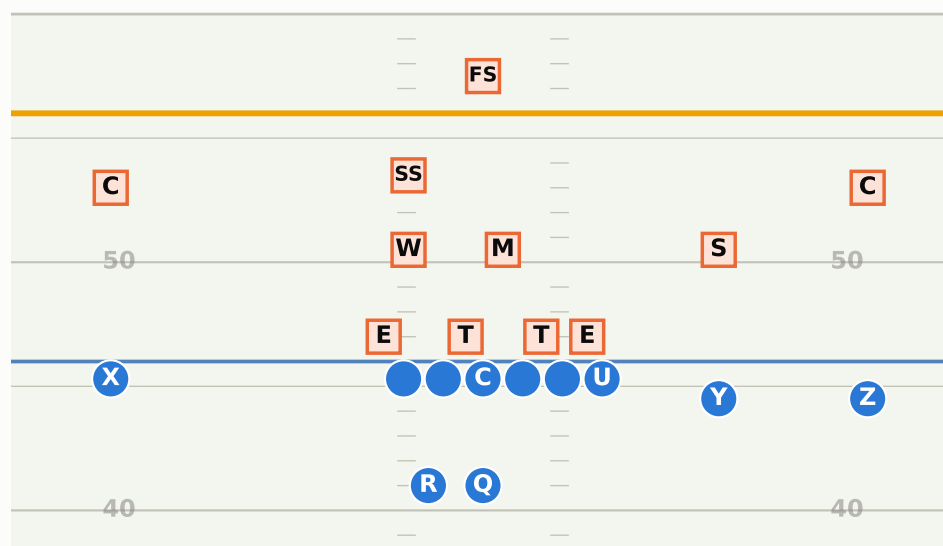


Figure 7: First-and-10 at the offense’s own 46. Between plays the offense sent a tight end on for a wide receiver, and the defense brought on a third linebacker. Now the offense lines up like this. Count the backs and the tight ends (the letters are your clue), then look at the right side.

What personnel is this? What did the defense choose, and what is the offense hoping for?

Answer 1 is at the end of the chapter. Try it first.

¹⁶Associated Press via Fox 2 Detroit, “Lions take issue with officials after potential winning 2-point pass is negated by penalty” (Dec. 31, 2023): <https://www.fox2detroit.com/news/lions-take-issue-with-officials-after-potential-winning-2-point-pass-is-negated-by-penalty> ; theScore, “Report: NFL won’t change eligible procedure after Cowboys-Lions controversy”: <https://thescore.com/news/2809328>

💡 Predict the play 2: the fullback is on the field

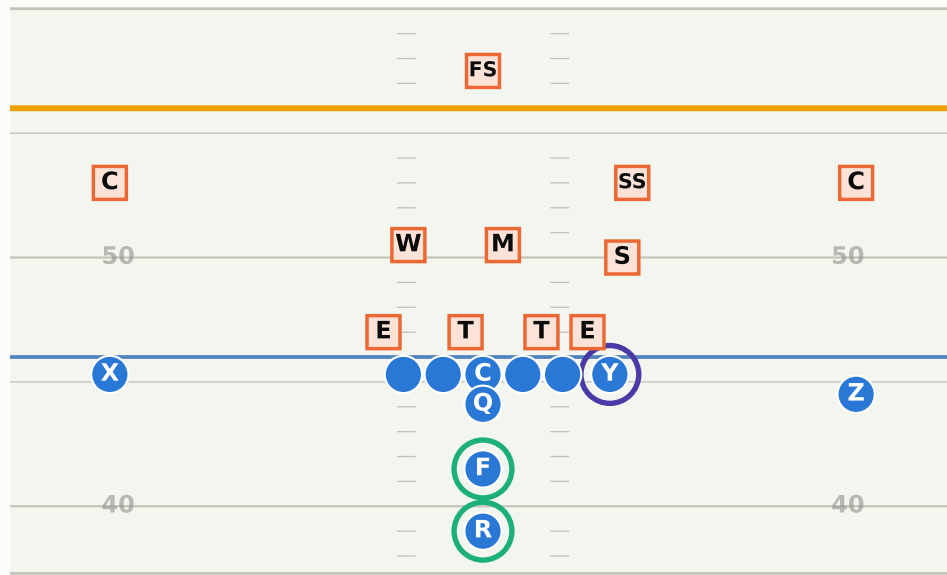


Figure 8: First-and-10, second quarter, tie game. The 49ers come out with a fullback (F) and a tailback (R) stacked behind the quarterback, one tight end and two wide receivers. The defense has matched with base.

This is 21 personnel. League-wide, should you lean run or pass? Does it matter that it's the 49ers?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: a lineman at tight end

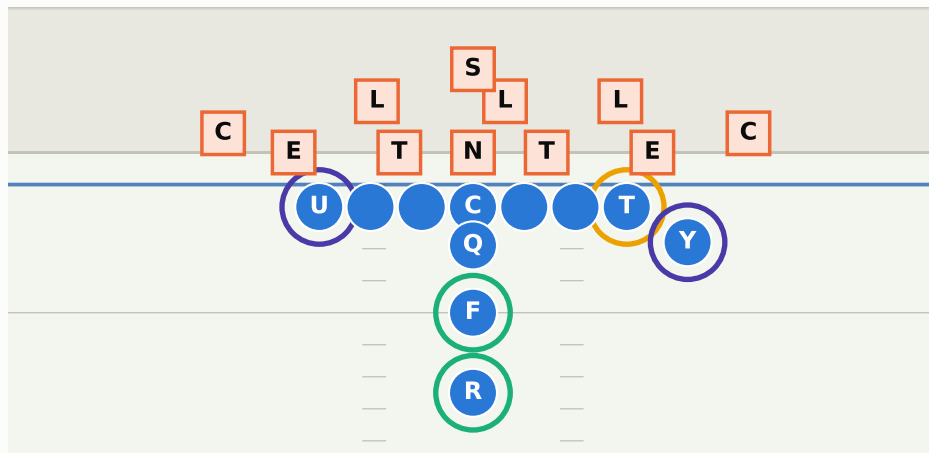


Figure 9: Second-and-goal from the 1. The offense brings on a sixth offensive lineman (T, gold ring) and lines him up at the right end of the line, where a tight end would stand. With him: a tight end on the left end (U), a tight end on the wing (Y), a fullback and a tailback. Before the snap, the referee announces: 'Number 70 has reported eligible.'

What grouping is this, what's the likely call, and who should the defense watch?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- A personnel grouping is two digits: **running backs, then tight ends**. Wide receivers are whatever is left of five ($WR = 5 - RB - TE$). A sixth lineman (jumbo) breaks the arithmetic.
- Personnel is the **first reveal** of every snap. It limits the offense's menu, and the defense sees it before the formation, which is why it's the first thing to count.
- The offense subs, then the **defense gets to match**: the umpire holds the snap until the defense has had its chance. An offense that keeps the same eleven gives the defense no such pause.
- Matching is a dilemma, not a formula. Versatile players (tight ends who catch, backs who split out) let the offense attack whichever choice the defense made.
- Personnel **tilts** run and pass but rarely decides it: on neutral early downs, 11 personnel passes a bit over half the time, 12 about half, 22 about a third. Read it with down, distance and the team's own habits.
- After a decade of 11 personnel, offenses went heavier in 2025 (12 and 13 at decade highs) to attack lighter defenses, and defenses began answering with more base: the arms race in miniature.

Connections

Builds on: The Twenty-Two (the positions being counted) and How to Watch a Broadcast Like You Mean It (the prediction log this chapter adds a column to). The radio, play clock and twelve-men penalty are from What a Play Really Is.

Sets up: Formation Rules and the Language of Alignment moves from *who* to *where*: legal formations, the X/Y/Z labels, and reporting eligible. Classic Formations and Spread Formations show what each grouping usually lines up in. A First Look at Defense builds out the defensive packages glossed here, and Motion and Shifts as Weapons shows how offenses use movement to expose the matchups personnel creates. Later: Box Math and Run Fits (why light boxes invite runs), Red Zone, Goal Line, and Short Yardage, and Tendencies, which builds team-by-team personnel scouting reports. Working with nflverse Data in Python rebuilds this chapter's charts step by step.

Case studies: the Shanahan tree (21 personnel and the fullback), the Ravens' run game, the Eagles in the trenches, and the Lions under Ben Johnson and Dan Campbell.

Go deeper

- Pat Kirwan, *Take Your Eye Off the Ball* (2010; 2nd ed. 2015): an NFL coach and scout's guide to the pre-snap picture, with a good chapter on personnel and substitutions.
- Chris B. Brown, *The Art of Smart Football* (2015): short essays on why offenses choose the players and formations they do.
- NFL Next Gen Stats' weekly "trend watch" pieces on NFL.com track personnel and defensive packages season by season.
- To count personnel yourself, nflverse's `load_participation()` has an `offense_personnel` string for every play since 2016; the folded code above parses both of its formats.

Answers to Predict the play

Answer 1.

12 personnel: one back (R) and two tight ends (U on the line, Y off the line in the slot), so two wide receivers. It *looks* like a three-receiver set with a tight end, which is the point.

The defense saw two tight ends run on and matched with **base**: three linebackers and four defensive backs. The offense then lined up so that Y, a tight end who can run routes, is split into the slot, and the defender over him is the Sam linebacker (S). That's the matchup from the dilemma figure: a linebacker trying to cover a receiver-like player in space. If the defense had stayed in nickel instead, the offense could have kept Y in tight and run at the lighter nickel back.

What to expect: a pass that isolates Y on the linebacker is a good bet, but not a certainty, since the offense may also like a run against a defense that had to widen a linebacker. What you've learned to look for is the *mismatch*, and the defense will try to fix it pre-snap, perhaps by sliding a safety down over Y or switching to man coverage with a safety on him. Motion and Shifts as Weapons shows how offenses use movement to force that decision into the open.

Answer 2.

League-wide, **lean run, gently**: on first and second down in neutral situations, 21 personnel passed about 45% of the time from 2023 to 2025. A fullback on the field is a real clue that the offense wants an extra blocker.

But it's the 49ers, and that changes the answer. San Francisco uses 21 personnel more than almost anyone, roughly a third of its snaps, and it **passes** from it more than the league does: about 53–54% on neutral early downs in each season from 2023 to 2025. Its fullback, Kyle Juszczyk, catches passes as well as he blocks, and play-action from this look is one of the offense's staples. The defense is in base because it respected the run, which leaves linebackers responsible for covering a fullback and a tight end. The best read is a coin flip with a play-action flavor.

The bigger lesson: personnel tendencies belong to teams, not just to groupings. Your prediction log will get sharper if you split it by team.

Answer 3.

It's **jumbo**: six offensive linemen, two tight ends and two backs, so **no wide receivers at all**. In nflverse's data it would appear as "6 OL, 2 RB, 2 TE, 0 WR" (22 by the digits). The likely call is a run: this is the heaviest look the offense has, at the spot where a yard is a touchdown, and in 2025 nearly a third of plays snapped from the 1- or 2-yard line had six or more linemen on the field.

But the referee's announcement matters. Number 70, the lineman at the right end of the line, reported eligible, so he is now a legal receiver, and the defense must cover him. Defenses facing heavy goal-line sets bring everyone close to the line to stop the run; the classic answer is a play-action pass to the player nobody is watching: a tight end leaking out, the fullback in the flat, or the big man who reported. If the defense loses track of number 70, he may be the target. If he *didn't* report and catches the ball, it's a penalty, as Detroit learned in Dallas in 2023.